PREP 5 · LESSONS 9–10

Half-life sums and staying safe

Counting half-lives · net decline as a ratio · contamination and irradiation

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 List alpha, beta and gamma radiation from most to least ionising. DN 271 [1]

Alpha, beta, gamma

2 Define half-life in terms of the number of nuclei. DN 637 [1]

The time taken for the number of nuclei of the isotope in a sample to halve.

3 State the two quantities that must balance in a nuclear equation. DN 727 [1]

The total mass number and the total atomic number

Key ideas LESSONS 9–10

Show your working

4 A sample has an activity of 640 Bq. Its half-life is 3 days.

a Calculate its activity after 9 days. [2]

9 ÷ 3 = 3 half-lives ✓

640 → 320 → 160 → 80 Bq ✓

b Calculate the net decline after 9 days, as a ratio of the initial activity. [1]

640 − 80 = 560 Bq; 560 : 640 = 7 : 8 ✓

5 What is radioactive contamination? [1]

The unwanted presence of radioactive atoms on or in other materials ✓

6 Give two precautions that protect people against irradiation. [1]

Any two: shielding, keeping a distance from the source, limiting the exposure time ✓

7 Explain why an alpha source is more dangerous inside the body than outside it. [2]

Outside, alpha is stopped by the skin (or air) ✓

inside, it is strongly ionising and is absorbed by nearby cells ✓